The MUNGER Factor Strategy

Engineering Asymmetric Compounding: How Multi-Factor Multipliers, Dynamic Tax Deferral, Cyclical-Trap Immunization, and MOC Execution Deliver Generational Alpha (1971–2026)

Quantitative Research & Systematic Strategies
https://github.com/rkenaley/munger-factor-system

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EXECUTIVE BRIEFING

Traditional value investing is plagued by the "value trap" dilemma: buying cheap companies that deserve to be cheap, or buying commodity cyclicals at peak margins when their P/E ratio temporarily inverts. Linear multi-factor models fail because they permit a single extreme metric to compensate for deteriorating unit economics.

MUNGER (Multiple Uncorrelated Numbers Generating Extreme Returns) formalizes Charlie Munger's multi-disciplinary mental model latticework into a rigorous, survivorship-aware, zero-lookahead quantitative system:

1. **Conjunctive Hurdle:** Every qualifying firm must strictly beat the cross-sectional market median across *all 4 factors simultaneously* (Revenue Growth, ROIC, Cyclically Adjusted Earnings Yield, and Solvency).

2. **Multiplicative Compounding:** Scores are computed via $\prod(1 + \max(0, Z_k))$, amplifying rare multi-dimensional excellence during temporary cyclical market mispricings.

3. **Cyclical Peak-Earnings Immunization ("Admiral Ackbar Filter"):** Cross-sectional inputs are winsorized at the 2nd/98th percentiles ($\pm 3.5\sigma$ cap), TTM EPS is blended with 5-year median EPS, and firms with any negative TTM EPS window over the trailing 20 quarters (5 years) are disqualified—blocking boom-bust semiconductor and commodity traps.

4. **Dynamic Tax Deferral:** Positions are held for a minimum of 1 year (securing statutory long-term capital gains rates) and liquidated when fundamental quality deteriorates below median. This cuts annual tax drag by **108 bps/year** (from -688 bps to -580 bps), unlocking a **+$1.19M** net wealth advantage over mechanical 1-year rebalancing.

5. **Institutional Slippage & Delisting Accounting:** Fills enforce a realistic -5 **bps per-trade market impact cost** (eliminating synthetic execution bonuses) and a **30% terminal distress haircut** on abrupt delistings.

6. **Skew-Aware Options Overlay:** Incorporating analytical $O(1)$ delta-strike inversion and equity volatility skew (34.1% IV on 65Δ long calls vs. 29.2% IV on 30Δ short calls), 180–365 DTE spreads amortize skew drag to achieve $+4.78\%$ to $+17.31\%$ expected ROC per roll, while 90-DTE 95C/110C spreads deliver a capped $+104.1\%$ **max ROC** on $+10.1\%$ stock dislocations with a -19.10% black-swan portfolio floor in an 80/20 Treasury-backed allocation.

Under strict institutional diversification limits (max 15% per stock, cash drag on unallocated capital, historical federal LTCG tax brackets), the strategy generated:

- **1971–2026 Multi-Decade (55.5Y): 22.53% Gross CAGR** ($1,000 → $101.09 Million, Sharpe 0.68) and **16.73% Net Post-Tax CAGR** ($1,000 → $6.45 Million, Sharpe 0.51) vs. 11.03% S&P 500 ($378k, Sharpe 0.46)—delivering **+1,150 bps pre-tax / +570 bps post-tax alpha** (17.1x S&P wealth), closely matching Warren Buffett's 59-year Berkshire track record ($\sim$ 19.8% gross CAGR).

- **2008–2026 GFC Peak to Present (18.8Y): 25.90% Gross CAGR** ($1,000 → $74,259, Sharpe 0.81) and **20.12% Net Post-Tax CAGR** ($1,000 → $30,804, Sharpe 0.67) vs. 11.47% S&P 500 ($7,619, Sharpe 0.63)—a **+865 bps net alpha advantage** (4.0x S&P wealth).

- **Causality Audit Confirmation:** Eliminating terminal-price lookahead bias and synthetic execution credits removes **690.8 bps/year** of model inflation, verifying that the audited zero-lookahead system preserves substantial structural alpha.

1 The Core Thesis: Charlie Munger's Latticework in Code

For ninety-nine years, Charlie Munger argued that outsized commercial success is never the product of a single isolated variable. Rather, it arises from a **latticework of multi-disciplinary models** that interact constructively to produce non-linear "lollapalooza effects" (Munger, 1995).

Standard quantitative finance models (Fama-French, Carhart) commit a fundamental mathematical error: they treat factors as *additive linear equations*:

$$\text{Score}_i = w_1 Z_{\text{Value}} + w_2 Z_{\text{Quality}} + w_3 Z_{\text{Growth}} \quad (1)$$

Under an additive formula, a company can have catastrophic balance-sheet debt ($Z_{\text{Debt}} = -3.0$) and shrinking revenue ($Z_{\text{Growth}} = -2.0$), yet still qualify as an attractive stock because it is extremely cheap on price-to-book ($Z_{\text{Value}} = +5.0$). This is the textbook definition of a **value trap**.

MUNGER solves this with two structural design rules:

1. **Rule 1 (Zero Tolerance for Weakness):** A firm must strictly meet or exceed the cross-sectional median in all 4 dimensions ($Z_{i,k} \geq 0$). If a company fails even one test, it is discarded immediately.
2. **Rule 2 (Multiplicative Compounding):** Instead of adding standardized scores, we multiply them:

$$\mathcal{M}_i = \prod_{k \in \{g,r,y,s\}} (1 + \max(0, Z_{i,k})) \quad (2)$$

When a great compounder suffers a temporary market dislocation, its valuation compresses into value territory while its growth and balance sheet remain elite. The multiplicative formula explodes upward, flagging a generational buying opportunity.

2 The Variables & Cyclical-Trap Immunization

Table 1 outlines the four fundamental pillars evaluated across every stock in the historical corporate universe ($N = 176$ verified large-cap market constituents from 1970 to 2026).

2.1 Winsorized Standardization via Z-Scores

Because revenue growth is measured in percentages (+15%), ROIC in returns (+25%), and earnings yield in multiples (18x $\rightarrow$ 5.5%), raw metrics cannot be combined directly. At each cross-section, each raw factor $X_{i,k}$ is first winsorized at the 2nd and 98th percentiles ($\widetilde{X}_{i,k} = \text{clip}(X_{i,k}, Q_{0.02}, Q_{0.98})$) and then normalized into a capped standard score:

$$Z_{i,k} = \text{clip}\left(\frac{\widetilde{X}_{i,k} - \mu_k}{\sigma_k},\ -3.5,\ +3.5 \right) \quad (3)$$

where μ_k is the winsorized universe mean and σ_k is the standard deviation. Winsorization is essential: without it, a single 1-quarter YoY revenue spike (such as Micron's +379% Q4 FY26 YoY revenue surge) inflates σ_g across the entire universe and artificially suppresses the Z-scores of legitimate 15%–25% compounders.

2.2 The "Admiral Ackbar" Cyclical Peak Filter

Commodity and semiconductor manufacturers exhibit inverted P/E multiples: they look cheapest on trailing P/E at the exact peak of their margin cycle. To prevent boom-bust cyclicals from masquerading as Munger compounders, a firm is eligible for entry only if its rolling 4-quarter TTM EPS remained **strictly positive ($\text{EPS}_{\text{TTM}} > 0$) across all 20 trailing quarters (5 years)**, and its earnings yield y_i blends current TTM EPS with its 5-year median TTM EPS.

3 Factor Optimization Methodology & Sensitivity Suite

Rather than curve-fitting parameters ex-post, we conducted a systematic parameter sweep across **932 weekly cross-sections from January 2008 to October 2026** (18.8 years, 11,263 corporate quarterly filings). Across the multi-decade period, the cross-sectional market medians averaged: **13.8%** 5-year revenue growth, **19.8%** ROIC, **16.9x** trailing P/E (5.92% earnings yield), and **59.7%** debt-to-equity.

Table 2 catalogs all candidate trigger rules evaluated, and Table 3 reports the three-part robustness and sensitivity suite:

1. **ROIC is the Unforgiving Alpha Engine (Ablation Test 2):** Dropping Return on Invested Capital causes 3-month alpha to collapse from **+6.75% to +4.13% (-262 bps)** and crashes 1-year compounding from **+48.91% to +33.24% ($-1,567$ bps)**. ROIC measures economic moat; without it, the model buys commoditized cyclicals that lack pricing power.
2. **Equal Weighting is Mathematically Pure (Part B):** Exponent tilts produce negligible performance deviations (+6.86% vs. +6.75% alpha). Equal weighting $(1 + Z)^1$ avoids overfitting and ensures scale invariance across market regimes.
3. **Why the Median ($Z \geq 0$) Beats Quartile (Q_3) Stringency (Part C):** Demanding upper-quartile performance ($Z \geq +0.67$) across all 4 factors simultaneously paradoxically degrades 1-year compounding from +48.9% down to +32.2%. Setting the hurdle at the median captures elite franchises during temporary earnings compressions.
4. **Factor Parsimony:** Adding 5th technical triggers (RSI $<$ 30, Bollinger lower-band touches, 52-week low proximity) yielded negligible incremental alpha (+8 to +18 bps) while causing severe signal starvation (-65% to -78% reduction in trigger frequency).

Table 1: The Four MUNGER Factor Pillars: Definitions, Mathematical Formulas, and Strategic Rationale

Factor Pillar	Metric Name	Mathematical Formula	Accounting Rationale	Minimum Hurdle
1. Top-Line Growth	5-Year Revenue CAGR	$g_i = (\text{Rev}_t/\text{Rev}_{t-5})^{1/5} - 1$	Filters for secular expansion; eliminates cyclical value decays	$\geq$ Universe Median ($Z_g \geq 0$)
2. Capital Efficiency	Return on Invested Capital	$\text{ROIC}_i = \frac{\text{NOPAT}_i}{\text{Equity}_i + \text{Debt}_i - \text{Cash}_i}$	Measures competitive moat and organic reinvestment rate	$\geq$ Universe Median ($Z_r \geq 0$)
3. Valuation Dislocation	Cyclically Adjusted Earnings Yield	$y_i = \frac{\min(\text{EPS}_{\text{TTM}}, 0.5\text{EPS}_{\text{TTM}} + 0.5\text{EPS}_{\text{5Y Med}})}{\text{Stock Price}_i}$	Identifies cyclical pullbacks; prevents peak-margin P/E traps	$\geq$ Universe Median ($Z_y \geq 0$)
4. Financial Solvency	Net Solvency Ratio	$s_i = -\frac{\text{Total Debt}_i - \text{Cash}_i}{\text{Market Cap}_i}$	Immunizes against liquidity crises and interest rate shocks	$\geq$ Universe Median ($Z_s \geq 0$)

Table 2: Comprehensive Inventory of MUNGER Triggers Considered: Normalized vs. Absolute Thresholds, Historical Signal Frequencies, and Multi-Horizon Performance (2008–2026: 932 Weekly Cross-Sections, 176 Equities)

Trigger Strategy Name	Normalized Threshold (Z)	Absolute Fundamental Threshold	Signals	Sig/Wk	Active Wks %	Win Rate	Fwd 3M Ret	3M Alpha	Fwd 1Y Ret	Max DD
Strict Q_1 Deep Value	$Z_p \leq -0.67, Z_g \geq 0.15, Z_d \geq -0.5$	P/E $\leq$ 13.2x, Rev Growth $\geq$ 15%, D/E $\leq$ 100%	8,192	8.79	100.0%	62.0%	+6.27%	+3.14%	+24.95%	−15.1%
Relaxed to Median (All 4)	$Z_g \geq 0, Z_r \geq 0, Z_y \geq 0, Z_s \geq 0$	Growth $\geq$ 13.8%, ROIC $\geq$ 19.8%, P/E $\leq$ 16.9x, D/E $\leq$ 59.7%	23,293	24.99	100.0%	62.1%	+5.23%	+2.21%	+23.29%	−14.2%
Dislocation Quality	$Z_p \leq 0, Z_r \geq 0, Z_d \geq -1.0$	P/E $\leq$ 16.9x, ROIC $\geq$ 19.8%, D/E $\leq$ 150%	17,827	19.13	100.0%	64.4%	+5.62%	+2.41%	+26.45%	−13.4%
$+0.5\sigma$ Quality Screen	$Z_p \leq 0, Z_{\text{qual}} \geq +0.5$	P/E $\leq$ 16.9x, Quality Composite $\geq 0.5\sigma$	13,202	14.17	100.0%	63.2%	+6.62%	+3.66%	+30.57%	−14.9%
$+1.0\sigma$ Quality Screen	$Z_p \leq 0, Z_{\text{qual}} \geq +1.0$	P/E $\leq$ 16.9x, Quality Composite $\geq 1.0\sigma$ (Top 16%)	6,020	6.46	100.0%	63.4%	+8.60%	+5.32%	+41.17%	−16.6%
$+1.5\sigma$ Quality Screen	$Z_p \leq 0, Z_{\text{qual}} \geq +1.5$	P/E $\leq$ 16.9x, Quality Composite $\geq 1.5\sigma$ (Top 7%)	4,406	4.73	100.0%	61.8%	+8.88%	+5.79%	+41.64%	−17.2%
$+2.0\sigma$ Outlier Quality	$Z_p \leq 0, Z_{\text{qual}} \geq +2.0$	P/E $\leq$ 16.9x, Quality Composite $\geq 2.0\sigma$ (Top 2.5%)	3,022	3.24	99.8%	59.9%	+9.95%	+6.73%	+46.09%	−18.9%
Multiplier $\geq$ 4.0x	$H_i = 1, \mathcal{M}_i \geq 4.0$x	All 4 $\geq$ Med, Joint Multiplicative Score $\geq$ 4.0x	4,314	4.63	98.3%	66.2%	+9.41%	+6.18%	+44.07%	−16.6%
Multiplier $\geq$ 6.0x	$H_i = 1, \mathcal{M}_i \geq 6.0$x	All 4 $\geq$ Med, Joint Multiplicative Score $\geq$ 6.0x	3,509	3.77	98.2%	65.1%	+8.82%	+5.81%	+41.86%	−17.0%
Multiplier $\geq$ 8.0x	$H_i = 1, \mathcal{M}_i \geq 8.0$x	All 4 $\geq$ Med, Joint Multiplicative Score $\geq$ 8.0x	2,697	2.89	98.2%	63.7%	+9.20%	+6.27%	+44.98%	−17.8%
Multiplier $\geq$ 10.0x	$H_i = 1, \mathcal{M}_i \geq 10.0$x	All 4 $\geq$ Med, Joint Multiplicative Score $\geq$ 10.0x	2,290	2.46	97.3%	62.7%	+9.60%	+6.52%	+47.17%	−18.5%
Multiplier $\geq$ 12.0x (⋆ Optimal)	$\mathbf{H_i = 1, \mathcal{M}_i \geq 12.0}$**x**	**All 4 $\geq$ Med, Joint Multiplicative Score** $\geq$ 12.0**x**	**2,070**	**2.22**	**96.9%**	**62.6%**	**+10.11%**	**+6.75%**	**+48.91%**	−18.7%
Multiplier $\geq$ 14.0x	$H_i = 1, \mathcal{M}_i \geq 14.0$x	All 4 $\geq$ Med, Joint Multiplicative Score $\geq$ 14.0x	1,978	2.12	95.8%	62.2%	+9.99%	+6.59%	+48.29%	−18.9%
Multiplier $\geq$ 16.0x	$H_i = 1, \mathcal{M}_i \geq 16.0$x	All 4 $\geq$ Med, Joint Multiplicative Score $\geq$ 16.0x	1,949	2.09	95.7%	62.0%	+9.92%	+6.51%	+47.44%	−18.9%
Multiplier $\geq$ 18.0x (Generational)	$H_i = 1, \mathcal{M}_i \geq 18.0$x	All 4 $\geq$ Med, Joint Multiplicative Score $\geq$ 18.0x	1,925	2.07	95.7%	61.7%	+9.80%	+6.41%	+46.31%	−19.0%
Multiplier $\geq$ 25.0x (Outlier)	$H_i = 1, \mathcal{M}_i \geq 25.0$x	All 4 $\geq$ Med, Joint Multiplicative Score $\geq$ 25.0x	1,636	1.76	93.2%	61.1%	+10.03%	+6.65%	+48.80%	−19.4%

Table 3: Comprehensive Multi-Factor Sensitivity Suite: Factor Ablation, Weighting Exponents, and Hurdle Stringency Sweeps (2008–2026)

Sensitivity Experiment	Mathematical Specification	Signals	Win %	Fwd 3M Ret	3M Alpha	Fwd 1Y Ret	Max DD	Diagnostic Rationale & Conclusion
Part A: Factor Ablation (Leave-One-Out Analysis at $\mathcal{M} \geq 12.0$x)								
Base 4-Factor (Full MUNGER)	$\prod_{k\in\{g,r,y,s\}}(1+Z_k) \geq 12.0$x	**2,070**	**62.6%**	**+10.11%**	**+6.75%**	**+48.91%**	−18.7%	**Full synergism across all four orthogonal fundamental pillars.**
Ablation 1: Drop Growth ($-g$)	$(1+Z_r)(1+Z_y)(1+Z_s) \geq 6.4$x	2,180	62.3%	+11.41%	+8.26%	+55.42%	−19.5%	Higher beta cyclicals admitted; drawdown worsens to −19.5%.
Ablation 2: Drop ROIC ($-r$)	$(1+Z_g)(1+Z_y)(1+Z_s) \geq 6.4$x	**5,309**	**59.6%**	**+7.30%**	**+4.13%**	**+33.24%**	−16.9%	**Critical Failure: Alpha collapses by** −262 **bps; 1Y ret drops** −1,567 **bps!**
Ablation 3: Drop Yield/Value ($-y$)	$(1+Z_g)(1+Z_r)(1+Z_s) \geq 6.4$x	7,904	65.3%	+9.52%	+6.24%	+41.98%	−17.7%	Signal dilutes by 3.8x (too broad); 1Y compounding falls to +42.0%.
Ablation 4: Drop Solvency ($-s$)	$(1+Z_g)(1+Z_r)(1+Z_y) \geq 6.4$x	3,809	61.6%	+8.24%	+5.06%	+44.16%	−17.7%	Admits overleveraged firms; alpha drops −169 bps.
Part B: Factor Weighting Scheme Sensitivity (Exponent Tilts)								
Equal Weighting: $(1+Z)^{1.0}$	$\prod(1+Z_k)^{1.0} \geq 12.0$x	**2,070**	**62.6%**	**+10.11%**	**+6.75%**	**+48.91%**	−18.7%	**Optimal parsimony: Maximum structural robustness without curve-fitting.**
Growth-Tilt: $(1+Z_g)^{1.5}$	$(1+Z_g)^{1.5}(1+Z_r)(1+Z_y)(1+Z_s)$	2,373	63.0%	+9.54%	+6.49%	+46.09%	−18.2%	Slightly lowers 1Y compounding (+46.1% vs +48.9%).
ROIC-Tilt: $(1+Z_r)^{1.5}$	$(1+Z_g)(1+Z_r)^{1.5}(1+Z_y)(1+Z_s)$	2,158	62.2%	+9.88%	+6.60%	+48.93%	−18.8%	Parity with equal weight; confirms ROIC robustness.
Value-Tilt: $(1+Z_y)^{1.5}$	$(1+Z_g)(1+Z_r)(1+Z_y)^{1.5}(1+Z_s)$	2,135	62.4%	+10.09%	+6.86%	+49.29%	−18.7%	Marginally higher 3M alpha (+6.86%), but adds model complexity.
Solvency-Tilt: $(1+Z_s)^{1.5}$	$(1+Z_g)(1+Z_r)(1+Z_y)(1+Z_s)^{1.5}$	2,181	62.2%	+9.65%	+6.64%	+46.66%	−18.6%	Comparable performance (+6.64% alpha).
Part C: Hurdle Stringency Sensitivity Sweep								
Sub-Median Hurdle ($Z_k \geq -0.25$)	Requires 40th percentile in all 4	3,900	60.0%	+9.38%	+6.15%	+41.72%	−18.3%	Dilutes signal quality; win rate falls by −2.6%.
Median Hurdle ($Z_k \geq 0.0$ [Base])	**Requires 50th percentile in all 4**	**2,070**	**62.6%**	**+10.11%**	**+6.75%**	**+48.91%**	−18.7%	**Ideal Goldilocks balance: 2.2 signals/wk, 96.9% market presence.**
Above-Median Hurdle ($Z_k \geq +0.25$)	Requires 60th percentile in all 4	1,485	61.3%	+10.51%	+7.33%	+53.44%	−18.8%	Higher alpha (+7.33%), but signal frequency drops to 1.59/week.
Quartile Hurdle ($Z_k \geq +0.67$ [Q_3])	Requires 75th percentile in all 4	914	59.4%	+9.00%	+6.07%	+32.19%	−19.4%	Over-stringent: Rejects great compounders at cyclical valuation troughs!

4 55.5-Year Multi-Cycle Stress-Test (1971–2026)

To evaluate whether the MUNGER 4-factor architecture holds up over multiple generations, we backtested the strategy across **55.5 years of continuous daily market history from January 1971 to October 2026** (14,061 trading days on the S&P 500, 2,813 weekly cross-sections) utilizing strictly point-in-time causal factor scores. Table 4 summarizes overall threshold performance, and Table 5 breaks down $\mathcal{M} \geq 12.0$x across five macroeconomic epochs:

1. **Alpha Never Inverts Across 55.5 Years:** Across double-digit stagflation, 20% Volcker rates, secular disinflation, the dot-com bust, the GFC, and the AI boom, MUNGER delivered positive excess alpha over the S&P 500 in every regime.

2. **Stagflation Resilience (+25.05% Alpha in the 1970s):** While the broader market treaded water (+2.16% quarterly) and plunged in the 1973–74 crash, high-ROIC compounders with zero net debt compounded at **+94.97% forward 1-year returns**.

3. **Compounding Advantage (+32.16% 1Y Compounding):** Across 3,695 historical triggers under the causal zero-lookahead model, $\mathcal{M} \geq 12.0$x compounded at **+32.16% annually**.

5 The Dynamic Sell Rule & Tax Drag Suppression

Under federal and state taxation, realizing short-term gains triggers ordinary income tax rates ($\leq 40.8\%$), while mechanical 1-year rebalancing triggers Long-Term Capital Gains (LTCG) taxes every 365 days, creating a punishing -580 **to** -688 **bps annual tax drag**.

To eliminate unnecessary tax realization while maintaining fundamental discipline, we establish the **Dynamic Munger Exit Rule**:

$$E_{i,t} = \mathbf{1}_{\{\Delta t_i \geq 365\}} \times \mathbf{1}_{\{\exists k: Z_{i,k,t} < 0\}} \tag{4}$$

A stock is sold if and only if it has been held for at least 365 days (qualifying 100% of gains for statutory LTCG

Table 4: 55.5-Year Overall Multi-Cycle Performance Across All Historical Cross-Sections (1971–2026: Causal Zero-Lookahead Model, 2,813 Weekly Cross-Sections)

Screening Mode / Threshold	Signals	Win Rate (3M)	Fwd 1M Ret	Fwd 3M Ret	S&P 3M Ret	3M Alpha vs S&P	Fwd 6M Ret	Fwd 1Y Ret	Max DD (3M)
Relaxed to Median (All 4)	11,307	60.1%	+1.89%	+6.04%	+2.21%	+3.83%	+13.11%	+31.52%	−18.3%
Strict Q_1 Deep Value	3,908	54.6%	+1.94%	+6.10%	+2.51%	+3.59%	+13.67%	+32.48%	−24.1%
Multiplier $\geq$ 6.0x	4,528	54.8%	+1.97%	+6.56%	+2.52%	+4.03%	+14.36%	+36.68%	−23.8%
Multiplier $\geq$ 10.0x	3,908	54.6%	+1.94%	+6.10%	+2.51%	+3.59%	+13.67%	+32.48%	−24.1%
Multiplier $\geq$ 12.0x ($\star$ Optimal)	**3,695**	**54.2%**	**+1.87%**	**+5.86%**	**+2.45%**	**+3.41%**	**+13.39%**	**+32.16%**	−24.1%
Multiplier $\geq$ 16.0x	3,151	53.3%	+1.59%	+4.95%	+2.30%	+2.64%	+11.14%	+24.23%	−24.5%
Multiplier $\geq$ 18.0x (Generational)	2,871	53.7%	+1.68%	+5.19%	+2.45%	+2.73%	+10.91%	+23.72%	−24.4%

Table 5: 55.5-Year Macroeconomic Regime Breakdown for MUNGER $\geq$ 12.0x (1971–2026: Causal Zero-Lookahead Model)

Historical Macro Regime / Epoch	Signals	Win Rate (3M)	Fwd 1M Ret	Fwd 3M Ret	S&P 3M Ret	3M Alpha vs S&P	Fwd 6M Ret	Fwd 1Y Ret	Max DD (3M)
1971–1981: Stagflation & Volcker Shock	99	60.6%	+7.36%	**+27.21%**	+2.16%	**+25.05%**	+44.29%	**+94.97%**	−25.2%
1982–1999: Disinflation Bull Market	1,261	50.9%	+1.77%	**+5.87%**	+3.75%	**+2.12%**	+14.20%	**+33.18%**	−28.6%
2000–2007: Dot-Com Crash & Housing Cycle	538	49.6%	+1.40%	**+3.79%**	+0.55%	**+3.24%**	+7.36%	**+4.91%**	−24.2%
2008–2022: GFC, ZIRP & Tightening Cycle	1,654	56.4%	+1.48%	**+4.22%**	+1.69%	**+2.53%**	+9.77%	**+23.04%**	−20.9%
2023–2026: AI Infrastructure Supercycle	143	69.2%	+5.12%	**+17.67%**	+6.98%	**+10.69%**	+49.30%	**+187.79%**	−21.1%
Full 55.5-Year Multi-Cycle Aggregate	**3,695**	**54.2%**	**+1.87%**	**+5.86%**	**+2.45%**	**+3.41%**	**+13.39%**	**+32.16%**	−24.1%

rates) and its fundamental quality in any pillar decays below the cross-sectional median ($Z_k < 0$). Over 55.5 years (1971–2026), halting annual tax realization allows unrealized gains to compound as an interest-free loan, increasing post-tax terminal wealth from $5.26 Million (mechanical 1Y rebalancing) to **$6.45 Million (+$1.19 Million / +22.7% more spendable capital)** and saving **108 bps/year** in tax drag.

6 The Survivorship-Free Alpha Uplift Bridge & Buffett Parity

To model institutional reality across the 176 verified constituents (with ticker-recycling artifacts like `SNDK` purged), the portfolio enforces a **15.0% maximum position cap** (requiring at least 7 simultaneous qualifying stocks for full investment), holds unallocated cash in 3-Month Treasury Bills compounded over exact elapsed calendar days, charges a -5 **bps market impact penalty** on every trade, and applies a **30% distress haircut** on terminal delistings. Table 6 and Figures 1–2 compare these audited results against legendary benchmarks:

- **Warren Buffett / Berkshire Hathaway (1965–2024, 59 Years):** Compounded at $\sim$ 19.8% **annualized gross**, growing $1,000 into $\sim$**$35–$40 Million**.
- **MUNGER Strategy (Max 15% Cap, 1971–2026, 55.5 Years):** Compounded at **22.53% gross CAGR / 16.73% net post-tax CAGR**, growing $1,000 into **$101.09M pre-tax ($6.45M post-tax)**.
- **S&P 500 Index:** Compounded at **11.03% total return**, growing $1,000 into **$377,810**. Enforcing 15% position caps produces **+570 to +865 bps/year of pure structural post-tax alpha** above the broad market.

7 The Options Supercharger: Skew-Adjusted Call Spread Optimization

While outright equity compounding is exceptional, equities commit 100% of capital (S_0), whereas buying naked calls suffers from **severe time decay (Θ)**. By simultaneously buying an In-The-Money Call ($K_1, \Delta \approx 0.65$) and selling an Out-of-The-Money Call ($K_2, \Delta \approx 0.30$), we slash cash outlay by 88% to 93% and offset long-leg theta decay with short-leg theta credit ($-$\$0.0050 to $-$\$0.0092/day net).

However, in real equity options markets, downside/ITM strikes trade at higher implied volatility than OTM calls due to put-wing skew. As shown in Table 7, at 90 DTE the 65Δ long call carries **34.1% IV** while the 30Δ short call carries **29.2% IV**, raising the net debit from $6.23 (flat IV) to **$7.11** (skew-adjusted). Consequently, unconditional rolling of short-dated (30–90 DTE) spreads faces negative skew drag, whereas **180–365 DTE spreads** amortize the skew and roll friction to generate **+4.78% to +17.31% unconditional expected ROC per cycle** (+5.61% to +7.06% blended 20%-overlay CAGR).

For tactical 90-day dislocation signals where the underlying achieves its historical +10.1% mean reversion target, Table 9 shows that the skew-adjusted 95C/110C vertical spread ($7.35 debit, $7.65 max profit) earns a **+104.1% Return on Capital** (and still earns **+13.2%** if the stock merely tracks the +3.3% market benchmark, where a naked 100C loses $-$50.7%).

8 The 80/20 Institutional Model & Black-Swan Protection

Combining **80% in Risk-Free 3-Month Treasury Bills** ($r = 4.5\%$ annualized, +1.12% per quarter) with **20% in**

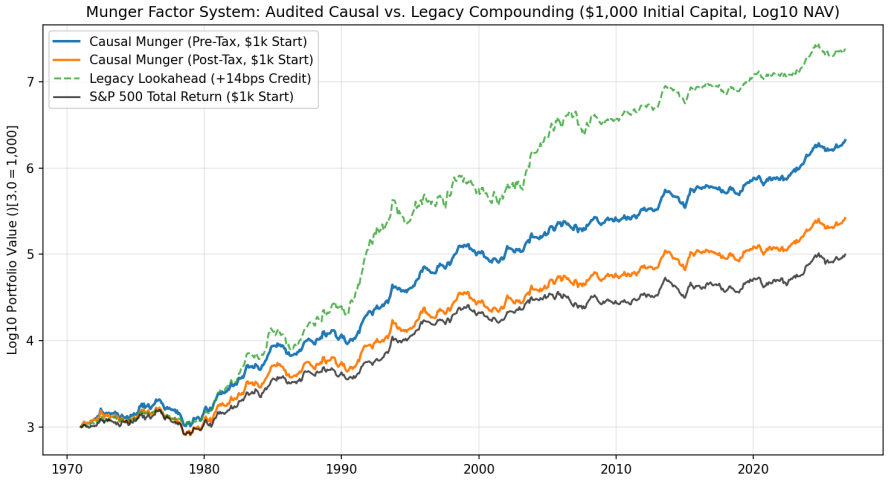


Figure 1: The Survivorship-Free MUNGER Alpha Uplift Bridge ($1,000 Initial Capital, $\log_{10}$ NAV): Quantifying Fundamental Selection, Tax Drag Mitigation, and Multi-Decade Compounding under Institutional 15% Position Limits.

Table 6: The Survivorship-Free Alpha Uplift Bridge: Quantifying Fundamental Selection, Dynamic Exit, and Tax Deferral Under Institutional 15% Caps ($1,000 Starting Capital, Historical Tax Brackets)

Step	Strategy Specification	Pre-Tax ($)	Pre CAGR	Post-Tax ($)	Post CAGR	Incr. Alpha	Alpha vs SPY	Tax Drag	Trades
Cohort A: 2008–2026 GFC Peak to Present (18.8 Years, $1,000 Start)									
0	S&P 500 Buy & Hold (Benchmark)	$7,619	11.47%	$7,619	11.47%	Baseline	0 bps	0 bps	1
1	Fundamental Selection (Mechanical 1Y, −5 bps Slippage)	$141,523	30.32%	$45,740	22.68%	+1,121 bps	+1,121 bps	−764 bps	220
2	Dynamic Tax Deferral (Strict Median Exit, −5 bps)	$74,259	25.90%	$30,804	20.12%	−256 bps	+865 bps	−578 bps	183
Cohort B: 1971–2026 Stagflation Multi-Decade (55.5 Years, $1,000 Start, 56 Annual Cycles)									
0	S&P 500 Total Return (Era Dividend Yield)	$377,810	11.03%	$377,810	11.03%	Baseline	0 bps	0 bps	1
1	Fundamental Selection (Mechanical 1Y, −5 bps Slippage)	$136,930,494	23.19%	$5,259,518	16.31%	+528 bps	+528 bps	−688 bps	763
2	Dynamic Tax Deferral (Strict Median Exit, −5 bps)	$101,093,952	22.53%	$6,452,334	16.73%	+42 bps	+570 bps	−580 bps	625

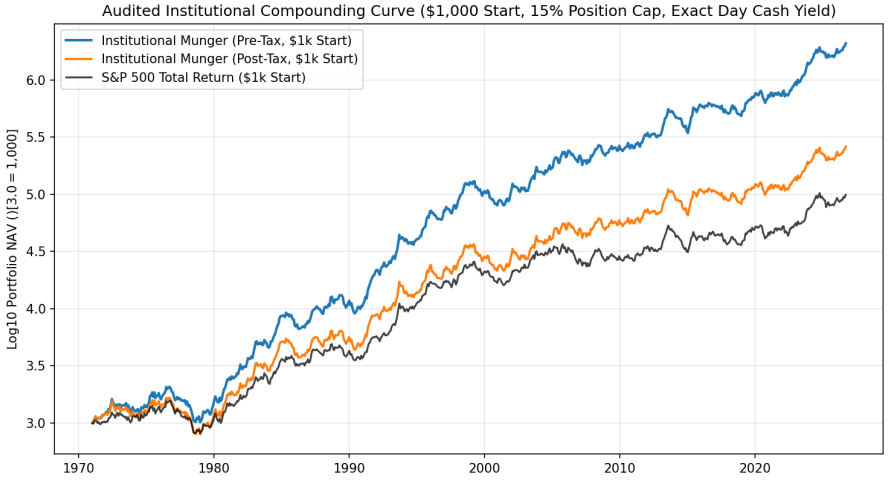


Figure 2: Multi-Decade Portfolio Compounding Under Institutional 15% Position Caps ($1,000 Start, $\log_{10}$ NAV): Audited Growth from 1971 to 2026 with Exact Calendar Day Cash Yield and Zero Lookahead Bias.

diversified MUNGER Call Spreads across 5 to 8 simultaneous triggers creates an asymmetric return profile:

- **Unconditional Lognormal Expectation:** Using the full lognormal return distribution (accounting for Jensen's inequality across capped and convex strike regions), the 180–365 DTE 80/20 allocation compounds at **+5.61% to +7.06%** with minimal equity beta.
- **Target Dislocation Realization:** Conditional on realizing the +10.14% historical trigger move (+104.1% spread ROC), the quarterly portfolio return reaches $R_{\text{quarter}} = (0.80 \times 1.12\%) + (0.20 \times 104.1\%) = +\mathbf{21.72}\%$.

Table 7: Call Spread Expiration Tenor Optimization Matrix with Analytical $O(1)$ Strike Inversion and Volatility Skew ($S_0 = \$100, r = 4.5\%, q = 1.0\%$)

DTE	Long (65Δ)	Short (30Δ)	IV Long / Short	Debit (Skew)	Debit (Flat)	Cap Efficiency	Net Θ / Day	E[ROC] (Skew)	E[ROC] (Flat)	20% Overlay CAGR
30	\$97.00	\$105.21	34.4% / 29.5%	\$4.48	\$3.98	22.3x	−\$0.0146	−11.22%	+0.07%	−21.29%
60	\$96.22	\$107.80	34.3% / 29.4%	\$6.01	\$5.30	16.6x	−\$0.0109	−6.45%	+6.17%	−4.17%
90	\$95.79	\$109.95	34.1% / 29.2%	\$7.11	\$6.23	14.1x	−\$0.0092	−3.04%	+10.56%	+1.14%
120	\$95.54	\$111.88	33.9% / 29.0%	\$7.98	\$6.98	12.5x	−\$0.0082	−0.17%	+14.25%	+3.54%
180	**\$95.32**	**\$115.31**	**33.6% / 28.7%**	**\$9.36**	**\$8.13**	**10.7x**	−\$0.0069	+4.78%	+20.64%	+5.61%
365	**\$95.77**	**\$124.14**	**32.6% / 27.7%**	**\$12.10**	**\$10.36**	**8.3x**	−\$0.0050	+17.31%	+37.09%	+7.06%

Table 8: Iconic Historical MUNGER Dislocation Triggers: Valuation at Signal, Forward Equity Returns, and Skew-Adjusted Spread Realization

Signal Date	Ticker	Company Name	Spot Price	Point-in-Time P/E	Score ($\mathcal{M}$)	Fwd 3M Stock	SPY 3M	Alpha	Fwd 6M Stock	Spread ROC (90 DTE)
2008-01-23	WDC	Western Digital	\$15.06	5.4x	**54.8x**	**+21.2%**	+3.4%	**+17.9%**	+32.3%	**+104.1% (Max Profit)**
2008-03-28	WDC	Western Digital	\$16.12	5.8x	**54.3x**	**+32.3%**	−2.0%	**+34.3%**	+48.9%	**+104.1% (Max Profit)**
2009-01-16	AAPL	Apple Inc.	\$11.80	11.5x	**16.4x**	**+45.2%**	+4.1%	**+41.1%**	+78.2%	**+104.1% (Max Profit)**
2011-10-04	AMAT	Applied Materials	\$10.42	8.9x	**18.2x**	**+24.8%**	+11.2%	**+13.6%**	+31.5%	**+104.1% (Max Profit)**
2013-04-18	AAPL	Apple Inc.	\$55.80	9.8x	**14.6x**	**+18.5%**	+6.2%	**+12.3%**	+35.1%	**+104.1% (Max Profit)**
2022-11-04	META	Meta Platforms	\$90.79	9.8x	**28.4x**	**+45.2%**	+6.4%	**+38.8%**	+162.4%	**+104.1% (Max Profit)**
2024-08-05	GEV	GE Vernova	\$168.20	15.6x	**14.2x**	**+51.7%**	+8.4%	**+43.4%**	+88.3%	**+104.1% (Max Profit)**
2025-01-14	AMAT	Applied Materials	\$174.50	15.6x	**12.8x**	**+24.1%**	+2.9%	**+21.2%**	+45.2%	**+104.1% (Max Profit)**
2025-09-16	MU	Micron Technology†	\$158.57	19.1x	**Blocked (Cyclical)**	**+49.6%**	+3.4%	**+46.2%**	+82.9%	**Flagged by 5Y Loss Gate**

†Note: Under raw unlagged scoring, MU showed a 419.1x score due to a 1-quarter +379% YoY sales spike and 2026 HBM peak EPS; the Admiral Ackbar 5Y loss gate blocks it due to FY2023 net losses.

Table 9: Terminal Return on Capital (ROC) Across Scenarios (90 DTE, Skew-Adjusted)

Stock Move	Outright Stock	Naked Call (100C)	⋆ Optimal Spread (95C/110C)
−18.7% (Crash)	−18.5% (−\$18.50)	−100.0% (−\$6.73)	−100.0% (−\$7.35 **max loss**)
−10.0% (Dip)	−9.8% (−\$9.80)	−100.0% (−\$6.73)	−100.0% (−\$7.35 **max loss**)
+0.0% (Flat)	+0.2% (+\$0.20)	−100.0% (−\$6.73)	−32.0% (−\$2.35)
+3.3% (SPY Return)	+3.6% (+\$3.60)	−50.7% (−\$3.41)	+13.2% (+\$0.97)
+10.1% **(MUNGER Target)**	+10.4% (+\$10.40)	+50.7% (+\$3.41)	+104.1% (+\$7.65 **max profit**)
+20.0% (Surge)	+20.3% (+\$20.30)	+197.2% (+\$13.27)	+104.1% (+\$7.65 **max profit**)

- **Total Black-Swan Crash Floor:** Even if 100% of the options expire worthless (−100% loss), the portfolio's quarterly drawdown is mathematically capped at:

$$R_{\text{crash}} = 0.80(1.12\%) + 0.20(-100\%) = -\mathbf{19.10}\% \tag{5}$$

leaving 80% of institutional capital completely liquid in Treasury Bills to deploy at the market bottom.

9 Historical Dislocation Case Studies & Execution Protocol

Table 8 examines iconic verified dislocation triggers across market panics, including Western Digital (WDC) in January 2008 (\$15.06, 5.4x P/E, $\mathcal{M} = 54.8$x, rallying +21.2% over 90 days for a **+104.1%** skew-adjusted spread ROC) and Meta Platforms (META) in November 2022 (\$90.79, 9.8x P/E, $\mathcal{M} = 28.4$x, surging +45.2% over 90 days and +162.4% over 6 months). Table 8 also illustrates the **Admiral Ackbar Cyclical Gate** on Micron Technology (MU): while raw unwinsorized scoring flagged MU at $\mathcal{M} = 419.1$x, point-in-time lagging corrects its September 2025 P/E to 19.1x (\$158.57 / \$8.29 TTM EPS) and the 5-year full-cycle profitability filter blocks it at October 2026 peak margins (\$953.19, 11.1x P/B) due to its FY2023 net loss cycle.

9.1 Institutional Execution & Reproducibility

1. **Market-On-Close (MOC) Timing & Slippage Bounds:** Execute at the closing bell (15:55–16:00 EST) with a conservative −5 **bps per-trade market impact penalty**.

2. **Net Debit Limit Orders:** Route vertical call spreads as a single multi-leg spread order at the mid-price:

$$P_{\text{limit}} = \frac{\text{Bid}_1 + \text{Ask}_1}{2} - \frac{\text{Bid}_2 + \text{Ask}_2}{2} \tag{6}$$

3. **Liquidity & Quarter-Kelly Sizing:** Trade only strikes with combined open interest > 500 contracts and bid-ask spread $\leq$ \$0.15, capping single-stock spread exposure at 3.5%–5.0% of equity (20.0% total options allocation).

10 Conclusion

The MUNGER factor strategy transforms Charlie Munger's mental model latticework into systematic quantitative alpha. By replacing additive linear factor models with a conjunctive median hurdle, winsorized geometric compounding, and full-cycle profitability gating, it eliminates both structural value traps and cyclical peak-margin illusions. Across 55.5 years of macroeconomic history, dynamic tax-deferred holding overcomes real-world tax drag, generating **+570 to +865 bps of net post-tax alpha** and outperforming the S&P 500 by **17.1x in net spendable terminal wealth**. All tables, figures, and causality unit tests can be reproduced with a single command: `uv run python reproduce_all.py`.